

Windows Event Log Forensics



CTF Challenge : Windows Event Log Forensics

Target File: challenge.evtx

System Investigated: Windows 10 (PC01.example.corp / MSEDGEWIN10)

Key Concepts & Definitions

Before the investigation, it is important to understand the "Language of Logs":

- Windows Event Logs (.evtx): These are the "black boxes" of a Windows system. Every time a user logs in, a file is deleted, or a script runs, Windows records it here. They are essential for reconstructing an attacker's timeline.
 - Emotet: A notorious "Banking Trojan" that evolved into a primary distributor for other malware (like Ransomware). It is famous for using highly obfuscated PowerShell scripts and Base64 encoding to bypass antivirus.
 - Event ID vs. Record ID: *Event ID*: The type* of action (e.g., ID 4104 always means a PowerShell script ran).
 - Record ID: The *serial number* of that specific log entry (e.g., the 27,736th thing that happened on the PC).
 - Ctrl + F (The Find Tool): In Event Viewer, this allows you to search for specific strings like "base64" or "net1.exe". *Pro-tip: Always click the first log in the list before searching so it searches from top to bottom.*
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Toolset Summary

Tool	Primary Use Case	How I used it
Event Viewer	Standard Windows log reader	Used to filter by Event IDs (400, 4104, 104).

Tool	Primary Use Case	How I used it
XML View	Deep dive into log metadata	Used to find hidden details like ProcessID and EventRecordID.
Filter Current Log	Narrowing down 77,000+ events	Isolated specific malicious activity out of thousands of normal logs.
Base64 Decoder	Reversing script obfuscation	Decoded the Emotet payload to see the actual attacker commands.

Investigation Steps

1 Detecting PowerShell Downgrade

I started by looking for signs of an attacker trying to bypass modern security by forcing an older version of PowerShell.

- Filter: Event ID 400
- Discovery: Found a session starting with HostVersion 2.0.
- Timestamp: 12/18/2020 6:50:33 PM

2 Anti-Forensics: Log Clearing

Attackers often "wipe the floor" after a crime. I searched for the specific event that records when logs are deleted.

- Filter: Event ID 104
- Evidence: The log was cleared. The Event Record ID was 27736.
- Machine Name: PC01.example.corp

3 Malware Analysis (Emotet Payload)

The most critical part of the investigation was finding the encoded malware script.

- Filter: Event ID 4104 (Script Block Logging).

- Discovery: Found a "Warning" log containing a massive \$base64data string.
- Process ID: 4836
- Timestamp: 12/13/2020 11:50:05 AM

Insider Threat (Privilege Enumeration)

Confirmed suspicions about an intern (IEUser) checking for administrative access.

- Filter: Event ID 4799.
 - Command: The intern used net1.exe to enumerate the Administrators group.
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Investigation Screenshots

◆ **Step 1: PowerShell Downgrade Found (ID 400)**

The screenshot shows the Windows Event Viewer interface. On the left, the 'Event Viewer (Local)' tree is expanded to 'Saved Logs', where 'challenge_1' is selected. The main pane displays a filtered list of events from the 'challenge_1' log. The filter is set to 'Log: file://C:\Users\abdu\OneDrive\Documents\challenge.evtx; Source: ; Event ID: 400. Number of events: 113'. The list shows five events, all with Level 'Information', Source 'PowerShell (PowerShell)', Event ID '400', and Task Category 'Engine Lifecycle'. The details pane for 'Event 400, PowerShell (PowerShell)' is open, showing the message 'Engine state is changed from None to Available.' and various details including 'NewEngineState=Available', 'PreviousEngineState=None', 'SequenceNumber=9', 'HostName=ConsoleHost', 'HostVersion=2.0', 'HostId=84a10cae-3d57-461c-8d55-994413a9c8bf', and 'EngineVersion=2.0'. Below the details, a summary section shows 'Log Name: Windows PowerShell', 'Source: PowerShell (PowerShell)', 'Event ID: 400', 'Level: Information', 'User: N/A', 'OpCode:', and 'More Information: [Event Log Online Help](#)'. The 'Logged' time is '12/18/2020 6:50:33 PM'. The 'Task Category' is 'Engine Lifecycle', 'Keywords' are 'Classic', and the 'Computer' is 'WIN-100UJBNP9G7'. On the right, the 'Actions' pane is visible, showing options like 'Open Saved Log...', 'Create Custom View...', 'Filter Current Log...', 'Clear Filter', 'Properties', 'Find...', 'Save Filtered Log File As...', 'Save Filter to Custom View...', 'View', 'Delete', 'Rename', 'Refresh', 'Help', 'Event Properties', 'Copy', 'Save Selected Events...', 'Refresh', and 'Help'. A watermark 'Activate Windows Go to Settings to activate Windows.' is visible in the bottom right corner.

Level	Date and Time	Source	Event ID	Task Category
Information	12/18/2020 6:50:33 PM	PowerShell (PowerShell)	400	Engine Lifecycle
Information	12/18/2020 6:48:45 PM	PowerShell (PowerShell)	400	Engine Lifecycle
Information	12/18/2020 6:48:45 PM	PowerShell (PowerShell)	400	Engine Lifecycle
Information	12/18/2020 6:48:43 PM	PowerShell (PowerShell)	400	Engine Lifecycle
Information	12/18/2020 6:48:42 PM	PowerShell (PowerShell)	400	Engine Lifecycle

Event 400, PowerShell (PowerShell)

General Details

Engine state is changed from None to Available.

Details:

NewEngineState=Available
PreviousEngineState=None

SequenceNumber=9

HostName=ConsoleHost
HostVersion=2.0
HostId=84a10cae-3d57-461c-8d55-994413a9c8bf
EngineVersion=2.0

Log Name: Windows PowerShell
Source: PowerShell (PowerShell) Logged: 12/18/2020 6:50:33 PM
Event ID: 400 Task Category: Engine Lifecycle
Level: Information Keywords: Classic
User: N/A Computer: WIN-100UJBNP9G7
OpCode:
More Information: [Event Log Online Help](#)

◆ Step 2: Log Clear Record ID (XML View)

Event Viewer (Local)

- Custom Views
- Windows Logs
 - Application
 - Security
 - Setup
 - System
 - Forwarded Events
- Applications and Services Logs
- Saved Logs
 - challenge
 - Microsoft-Windows-App
 - Microsoft-Windows-App
 - Microsoft-Windows-Cloud
 - challenge_1
- Subscriptions

challenge_1 Number of events: 77,524

Filtered: Log: file://C:\Users\abdu\OneDrive\Documents\challenge.evbx; Source: ; Event ID: 104. Number of events: 1

Level	Date and Time	Source	Event ID	Task Category
Information	3/20/2019 2:34:25 AM	Eventlog	104	Log clear

Event 104, Eventlog

General Details

☐ Friendly View ☒ XML View

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<Keywords>0x8000000000000000</Keywords>
<TimeCreated SystemTime="2019-03-19T23:34:25.8943413Z" />
<EventRecordID>27736</EventRecordID>
<Correlation />
<Execution ProcessID="812" ThreadID="3916" />
<Channel>System</Channel>
<Computer>PC01.example.corp</Computer>
<Security UserID="S-1-5-21-1587066498-1489273250-1035260531-1106" />
</System>
- <UserData>
  
```

Activate Windows
Go to Settings to activate Windows.

ites a filter.

◆ Step 3: Emotet Encoded Payload & Attack Date

File Action View Help

Event Viewer (Local)

- Custom Views
- Windows Logs
- Applications and Services Logs
- Saved Logs
 - challenge
 - challenge1
 - challenge_1
 - Microsoft-AppV-Client%
 - Microsoft-Windows-App
 - Microsoft-Windows-App
 - Microsoft-Windows-Clo
 - Application
 - Subscriptions

challenge_1 Number of events: 77,524

Filtered: Log: file://C:\Users\abdu\OneDrive\Documents\challenge.evtx; Source: ; Event ID: 4104. Number of events: 678

Level	Date and Time	Source	Event ID	Task Category
Verbose	8/26/2020 8:09:28 AM	PowerShell (Microsoft-...	4104	Execute a Remote Com...
Verbose	12/13/2020 11:49:25 AM	PowerShell (Microsoft-...	4104	Execute a Remote Com...
Verbose	12/13/2020 11:49:25 AM	PowerShell (Microsoft-...	4104	Execute a Remote Com...
Verbose	12/13/2020 11:49:25 AM	PowerShell (Microsoft-...	4104	Execute a Remote Com...
Verbose	12/13/2020 11:49:25 AM	PowerShell (Microsoft-...	4104	Execute a Remote Com...
Verbose	12/13/2020 11:49:25 AM	PowerShell (Microsoft-...	4104	Execute a Remote Com...
Warning	12/13/2020 11:50:05 AM	PowerShell (Microsoft-...	4104	Execute a Remote Com...

Event 4104, PowerShell (Microsoft-Windows-PowerShell)

General Details

Creating Scriptblock text (1 of 1):
\$base64data = "H4sIAIGp1V8CA51WbW/bNhD+719xcLVaQixCdlt5DZCirrJuAbLWqLzlg2EgtHSOtcikR1J+QeL/XIKILDIO0Kb6You8e+65545HvYihX6OY5Qx8uBapUshguoVP+meUC4YCXsM5XSH8RUWyybbW0ZaxSzuBPVP41TuMsRaagdd8C/TjrGM7gC679r9P/MFbgj7ZL/EIXqBcV0fZhYV8Zk38knuOM5pkKBSZ6J6WZ1BCOEjnrYaCb7bkYVeb6xUtq1dTXFZpda6h2J/SAVduOX/caREym4nTsgXC8qS7uFqJLOys0eL53zNMk6TYtWzmlLHKCVYARY8yTMOBD+6HpQm6QzcKgZ+D+0pylL2l6xWfoVvIkqtfxa8jMdcqv/L4hRLeLxHSpJRvHyypMer0/fn977EmkokKZwDZ0sWtrdNawG8QxLpVGL0vhlx2z/EVUElh8ZjyHrpR86eoh0MbqN0LSO/Ne9lL3pD+23ftbpGJD8qNZRKIF0YviU80b0WFWuaZ82wLFFJ0LR121akQU/KLKrAnmGlca7bfkuiytS18bvOTPCvdt17Z6TRd+BTCeMDn2+44ApDFCqdpTFV+ C/N0oSa5gtplk1pfDfxcFokEGu5qZzjdNAPqON16hhrUmdU1O08XSrcDyZOOBxtF9ASD/Qz8Nv98HO6oosqbbdscKNishinpjePj0dROHlpWle0/mRs3Pa1bIK+luWEiOaYZSByxrQ1aCVyqRu1DSfqfFudmjdmjvmJXtNF2W/EfLHMVb15w0K+

Log Name: Microsoft-Windows-PowerShell/Operational
Source: PowerShell (Microsoft-Wind
Event ID: 4104
Level: Warning
User: S-1-5-21-1881654409-284601
OpCode: On create calls
More Information: [Event Log Online Help](#)

Task Category: Execute a Remote Command
Keywords: None
Computer: WIN-100UJBNP9G7

Activated Windows
Go to Settings to activate Windows.

Execution Process ID

Event Viewer

File Action View Help

challenge_1 Number of events: 77,524

Filtered Log: file://C:\Users\abdu\OneDrive\Documents\challenge.evtx; Source: ; Event ID: 4104. Number of events: 678

Level	Date and Time	Source	Event ID	Task Category
Verbose	8/26/2020 8:09:28 AM	PowerShell (Microsoft-...	4104	Execute a Remote Com...
Verbose	12/13/2020 11:49:25 AM	PowerShell (Microsoft-...	4104	Execute a Remote Com...
Verbose	12/13/2020 11:49:25 AM	PowerShell (Microsoft-...	4104	Execute a Remote Com...
Verbose	12/13/2020 11:49:25 AM	PowerShell (Microsoft-...	4104	Execute a Remote Com...
Verbose	12/13/2020 11:49:25 AM	PowerShell (Microsoft-...	4104	Execute a Remote Com...
Verbose	12/13/2020 11:49:25 AM	PowerShell (Microsoft-...	4104	Execute a Remote Com...
Warning	12/13/2020 11:50:05 AM	PowerShell (Microsoft-...	4104	Execute a Remote Com...

Event 4104, PowerShell (Microsoft-Windows-PowerShell)

General Details

☐ Friendly View ☒ XML View

```

<EventID>4104</EventID>
<Version>1</Version>
<Level>3</Level>
<Task>2</Task>
<Opcode>15</Opcode>
<Keywords>0x0</Keywords>
<TimeCreated SystemTime="2020-12-13T08:50:05.2765640Z" />
<EventRecordID>3766</EventRecordID>
<Correlation ActivityID="{f305b354-cf12-0000-99b3-06f312cfd601}" />
<Execution ProcessID="4836" ThreadID="520" />
<Channel>Microsoft-Windows-PowerShell/Operational</Channel>
<Computer>WIN-100UJBNP9G7</Computer>
<Security UserID="S-1-5-21-1881654409-284601696-3713096779-500" />
</System>
- <EventData>
  <Data Name="..." />

```

Actions

challenge_1

- Open Saved Log...
- Create Custom View...
- Import Custom View...
- Filter Current Log...
- Clear Filter
- Properties
- Find...
- Save Filtered Log File As...
- Save Filter to Custom View...
- View
- Delete
- Rename
- Refresh
- Help
- Event 4104, PowerShell (Microsoft-Win...
- Event Properties
- Copy
- Save Selected Events...
- Refresh
- Help

Activate Windows
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◆ Step 4: Intern Group Enumeration

Event Viewer

File Action View Help

Event Viewer (Local)

- Custom Views
- Windows Logs
- Applications and Services Logs
- Saved Logs
 - challenge
 - challenge1
 - challenge_1
 - Microsoft-AppV-Client%
 - Microsoft-Windows-App
 - Microsoft-Windows-App
 - Microsoft-Windows-Clou
 - Application Subscriptions

challenge Number of events: 77,524

Level	Date and Time	Source	Event ID	Task Category
Information	12/5/2020 3:54:40 AM	EventLog	6005	None
Information	12/5/2020 3:54:40 AM	EventLog	6009	None
Information	8/26/2020 8:09:33 AM	PowerShell (PowerShell)	800	Pipeline Execution Details
Verbose	8/26/2020 8:09:28 AM	PowerShell (Microsoft-...	4104	Execute a Remote Com...
Information	8/5/2019 12:25:03 PM	Microsoft Windows sec...	4799	Security Group Manage...
Information	8/5/2019 12:24:56 PM	Microsoft Windows sec...	4798	User Account Managem...
Information	8/5/2019 12:24:56 PM	Microsoft Windows sec...	4798	User Account Managem...
Information	3/20/2019 2:34:25 AM	Eventlog	104	Log clear

Event 4799, Microsoft Windows security auditing.

General Details

Security ID: S-1-5-21-3461203602-4096304019-2269080069-1000
 Account Name: IEUser
 Account Domain: MSEDGEWIN10
 Logon ID: 0x2E47A

Group: Security ID: BUILTIN\Administrators
 Group Name: Administrators
 Group Domain: Builtin

Log Name: Security
 Source: Microsoft Windows security
 Event ID: 4799
 Level: Information
 User: N/A
 OpCode: Info
 More Information: [Event Log Online Help](#)

Logged: 8/5/2019 12:25:03 PM
 Task Category: Security Group Management
 Keywords: Audit Success
 Computer: MSEDGEWIN10

Actions

- challenge
- Open Saved Log...
- Create Custom View...
- Import Custom View...
- Filter Current Log...
- Properties
- Find...
- Save All Events As...
- View
- Delete
- Rename
- Refresh
- Help
- Event 4799, Microsoft Windows security...
- Event Properties
- Copy
- Save Selected Events...
- Refresh
- Help

Activate Windows
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Captured Forensic Flags

- Downgrade Event ID: 400
- Log Clear Record ID: 27736
- Emotet Process ID: 4836

- Suspicious User: IEUser
- Target Group: Administrators

Created by Abdu - 2026